Prob.1. (2&T, Ch. 7, 8.7)

Suppose that f is defined & cont-s on $[-\pi, \pi]$. State the conditions under which the F.S of f converges to $f(x)$ for <u>every</u> $x \in [-\pi, \pi]$.

Sol-n:

Recall the following properties of Fourier series:

(i) F-S is periodic on $[-\pi, \pi]$.

(ii) What f-ns have the F-S? Answer: ALL integrals that are finite:

$$|a_n \cos(n\theta) + b_n \sin(n\theta)| \leq |a_n| + |b_n|$$

(iii) When does the F-S converge to its f-n $f(x)$?

If $f(x)$ is <u>piecewise smooth</u> on $-\pi \leq x \leq \pi$;

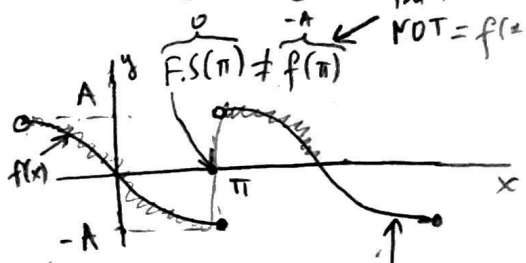
<u>Piecewise smooth</u>: The interval of interest can be broken up to pieces where $f(x) \in C^1(I_i)$ where $\cup I_i = \Omega$.

But the F.S converging f-n is NOT $= f(x)$

(iv) <u>Matching end points</u>, that is

$$f(-\pi) = f(\pi)$$

e.g.



This is the f-n that F-S converges to.

(iii) answers the question, (iv) can be extra supporting.

Prob.2. (2&T, Ch. 7, 8.10)

Derive the F-S for the ↓ f-ns & then sketch the graph of the f-n to which the series converges for all x.

(a)

$$f(x) = \begin{cases} 0, & \text{for } -\pi < x < 0 \\ 1, & \text{for } 0 < x < \pi \end{cases}$$

sol-n

[Preamble]